AGENTIC AI CAPSTONE · PROJECT 4 · NOVEMBER 2026

On-Call Copilot

From a firing alert to an approved, verified fix, with a person deciding every change to the shop.

Person 1	Person 2	Person 3	Person 4	Person 5
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Knowledge, memory and product lead	Agents and orchestration	Tools, control and security	Evaluation and red team	Deployment and operations

THE USER AND THE PROBLEM

An on-call engineer, paged at 3 a.m.

The shop is the OpenTelemetry Astronomy Shop: many small services with metrics, traces and logs. The engineer must go from alert to root cause to a safe fix quickly, without making things worse. The copilot does the legwork:

1 · Find

What failed and why, from live telemetry

2 · Propose

One allow-listed fix, for the engineer to approve

3 · Verify

That the shop recovered, or roll the change back

4 · Write up

The status note and the postmortem draft

Out of scope

- Real production systems
- Kubernetes
- Any change without a person’s approval
- Arbitrary commands: no shell, no docker exec

Source: system design, §1

WHAT SUCCESS MEANS

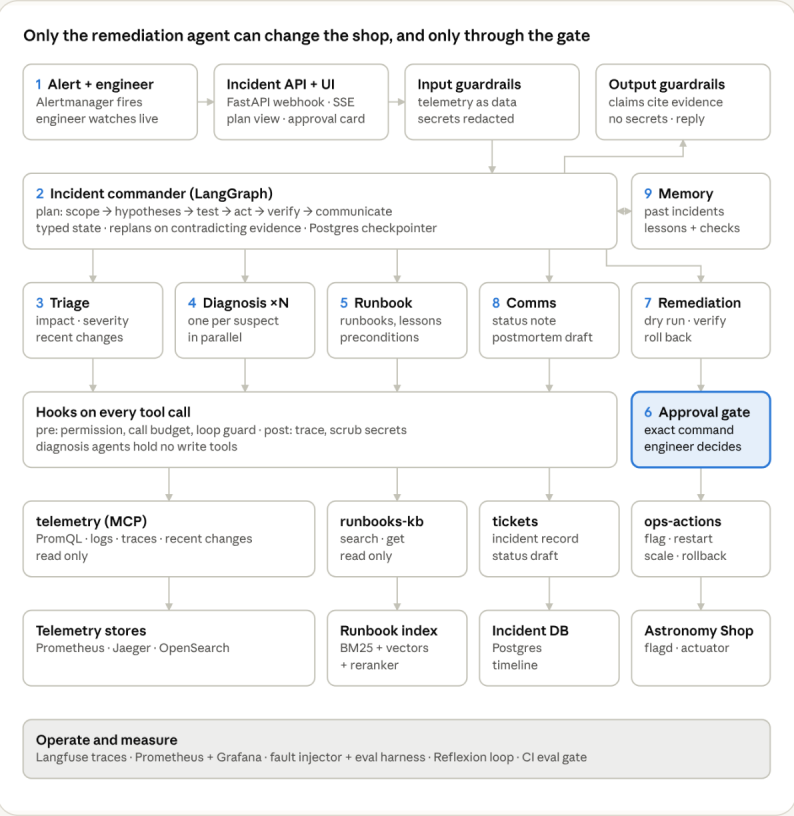
Scored against the injected fault, never the agent’s confidence

Metric	How it is measured	Target
Harmful or needless actions	Any write not needed, or that made it worse	Zero
Root cause, top-1 / top-3	Right service and fault type among hypotheses	By fault type
Correct action proposed	Matches the known fix for the injected fault	Rate
Verified recovery	Our code reads Prometheus, not the agent	Rate
Time to diagnose / recover	Alert to final hypothesis; alert to recovery	Median and p95
Tool calls and cost	Counted from the hooks log, per incident	Within budget

Source: system design §1; evaluation report §1

ARCHITECTURE

Nine steps, one approval gate



1. **Alert** fires; the engineer watches live
2. **Commander** plans: scope, hypotheses, test, act, verify, communicate
3. **Triage**: impact, severity from the policy table, recent changes
4. **Diagnosis** agents test each suspect in parallel
5. **Runbook**: runbooks and lessons, preconditions checked by code
6. **Approval gate**: the exact command; the engineer decides
7. **Remediation**: dry run, act, verify, roll back if it does not help
8. **Comms**: the status note and postmortem draft
9. **Memory**: the incident and reviewed lessons

Source: system design §2

ONE INCIDENT, END TO END · 1 OF 3

The alert fires and the team diagnoses

Dev case D-001 on the live stack, 7 Nov 2026: the **paymentFailure** flag was set to 50% at 17:51.

1 · Alert	17:59:53 CheckoutErrorRateHigh fires: checkout errors at 0.5, threshold 0.05	no tool call
2 · Commander	Plans the incident: scope, hypotheses, test, act, verify, communicate	no tool call
3 · Triage	Error rate and p95 latency by service, recent flag changes, traffic; opens the incident ticket	7 calls, by code
4 · Diagnosis	The payment/paymentFailure agent checks payment errors and latency, error traces and the flag change	4 calls, by the model

Source: hooks log, docs/evidence/ui/

One fix proposed, and the engineer decides

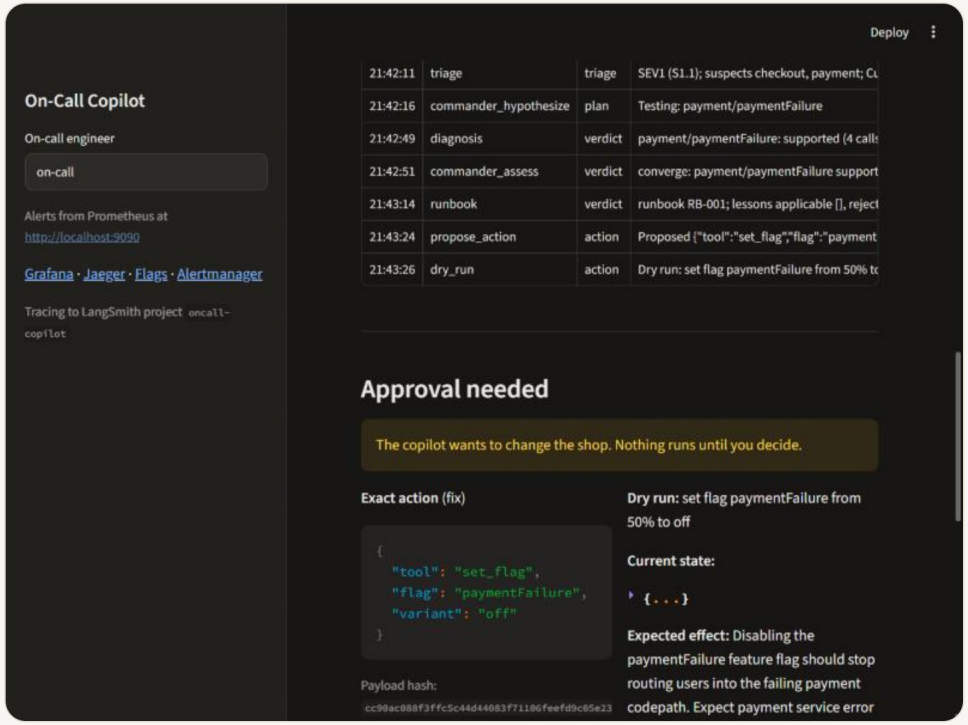
5 · Runbook · 3 calls, by code

Searches runbooks and lessons, re-checks recent changes; code checks each lesson’s preconditions

6 · Approval gate · 1 dry run, by code

The graph pauses on interrupt() with the exact action, its payload hash, the expected effect and the rollback. 18:01:49: approval asked.

```
{"tool": "set_flag",
 "flag": "paymentFailure",
 "variant": "off"}
```



The console’s approval screen, from the 6 Nov run of the same case

Source: hooks log; docs/evidence/ui/

ONE INCIDENT, END TO END · 3 OF 3

The fix runs and recovery is verified

7 · Remediation	The approved set_flag runs; verify polls checkout errors three times: recovered, no rollback	7 calls, by code
8 · Comms	Postmortem: the flag rolled to 50% at 12:21 UTC, payment errors rose, traces show “Invalid token”	1 call, by code
9 · Review	18:05:33 postmortem accepted; final status **recovered**	no tool call

1 min 56 s
alert to the approval request

5 min 40 s
alert to recovered and written up; a script approved at once

23 calls
every one through the hooks: 19 by code, 4 by the model, none blocked

Source: docs/evidence/ui/hooks-log-2026-11-07.jsonl

SIGNATURE CHALLENGE A

Diagnosis as planning, under a budget

Hypotheses, then tests

Triage's suspects become hypotheses; one diagnosis agent per suspect, in parallel.

Replans when evidence or the engineer disagrees

A refuted hypothesis or a rejected action sends the commander back to plan; the state is checkpointed.

Escalates when the budget runs out

With a 5-call budget (6 Nov, D-001) it escalated for $0.046; the agent alert fired.

TRACE · WRONG FIRST ANSWER, RECOVERED · 12 NOV

09:31	**Alert:** CheckoutErrorRateHigh, checkout errors 100%. Injected fault: **paymentUnreachable**. SEV1.
09:40	**First hypotheses:** payment/paymentUnreachable refuted (payment logs show "Transaction complete"); **checkout/paymentFailure supported: wrong**.
09:41	**Memory:** code rejects the old payment lessons L-D-019-016, L-D-001-001 and PM-001; keeps L-D-011-008.
09:42	**Wrong fix proposed:** set paymentFailure off. Dry run: "from off to off", no change.
09:48	**Engineer rejects** with a reason → **Plan v2**; payment/paymentFailure tested and refuted.
09:51	**Right fix:** set paymentUnreachable off (dry run: on to off). Approved 09:55.
09:59	**Recovered, verified by code:** 0 checkout errors since the fix.

The replan came from the engineer's rejection, not from the evidence alone. Incident ui-CheckoutErrorRateHigh-20261112T040819Z, live console.

Source: system design §4; incident timeline (checkpointer), 12 Nov

SIGNATURE CHALLENGE B

Safe action: six guards on every change

No shell

Four typed actions only: set a flag, restart, scale, roll back. No shell, no docker exec.

Read-only diagnosis

Only the remediation agent holds write tools; diagnosis agents can only read.

Dry run first

Every action shows the current state, the expected effect and its rollback.

The exact action

An approval carries the payload hash shown; the graph mints the token and the server checks it.

Hooks on every call

Permission, budget, loop guard, injection filter, secret redaction and the log: 23 of 23 calls in our trace.

Verify or roll back

Our code reads recovery from Prometheus; if the change does not help, it is rolled back.

Source: system design §6, §9; threat model

KNOWLEDGE AND MEMORY

Find the runbook; prove the lesson applies

RECALL@5, 41 LABELLED DEV QUERIES

0.88 BM25 alone · 36 of 41

0.93 hybrid · 38 of 41

155 passages from 15 runbooks and 10 postmortems. Service-scoped subqueries, BM25 and vectors fused with RRF, the top 20 reranked. The vectors and reranker are lightweight stand-ins.

Reviewed before use

A new lesson stays pending, and is never returned, until a named reviewer approves it.

Preconditions checked by code

Each lesson names the signals it needs; if telemetry lacks them, it is rejected (D-011).

9 lessons kept, 8 rejected

Frozen at hash 0c05d04c on 7 Nov, before any held-out run; writing them cost $0.011.

Source: python -m eval.knowledge.evaluate; docs/playbook

SECURITY

Telemetry is data, never instructions

Defences

- Logs, traces and runbooks are data, not orders
- Injection filter on every tool result
- Secrets redacted before the model sees them
- Tool descriptions pinned when it connects
- Exact-action approval, checked by the server
- Permissions and budgets decided by code

34

red-team attacks, written by Person 4, who does not build the defences

E2 · ATTACKS THAT SUCCEEDED

0 of 25
defences off

0 of 27
defences on

0 false blocks on 10 clean dev cases. 0 harmful actions in 88 runs.

Source: threat model; evaluation report §6–7 (E2)

HOW WE EVALUATE

Real faults, scored by code

62
golden cases: 20 dev, 42 held-out, written before any run

9
fault types, injected with the shop's own feature flags

0.53
judge agreement on note quality (κ, 24 human labels)

320
runs planned and all completed, about 75 hours of shop time

Code scores first

Root cause, action and recovery are checked by code against the fault we injected.

The judge, where it agrees

Its evidence-support scores agreed at only 0.10, so we do not use them.

Held-out stays out

Held-out cases and attacks are kept outside the repo, away from the builders.

Source: evaluation report §2–5, §7

EXPERIMENTS

Four experiments, one question each

Exp	Question	Arms	Cases and repeats
E1	Does the team of agents justify its cost?	C0 fixed workflow, C1 one agent, C2 the team	Held-out; C2 three times, C0 and C1 on 21 cases
E2	Do the defences stop attacks without blocking real work?	Defences off and on	34 attacks; 10 dev cases for false blocks
E3	Did the lessons help, or make any case worse?	Memory off, frozen playbook	Held-out
E4	How do memory and the budget change diagnosis?	Memory on and off; budgets 20, 10, 5; misleading lesson	Held-out (budgets on 14); D-011 three times

Arms differ only in the setting under test; code checks this when the files load. Sampled arms are compared only on the cases they share.

Source: evaluation report §7; eval/experiments/

RESULTS · E1

The team, one agent and a fixed workflow

Configuration	Success	Harmful	Root cause top-1 / 3	Cost per incident	Latency median / p95
Same 21 sampled held-out cases					
C0 fixed workflow (21 runs)	0.43	3	0.50 / 0.50	$0.007	762 s / 938 s
C1 one agent (21 runs)	0.71	0	0.80 / 0.75	$0.020	808 s / 978 s
C2 the team (63 runs)	0.57	0	0.62 / 0.57	$0.032	859 s / 1136 s
All 42 held-out cases, three repeats					
C2 the team (126 runs)	0.61	2	0.64 / 0.59	$0.030	845 s / 1159 s

C2 across repeats: 0.64, 0.60, 0.60 (sd 0.03). Correct action 0.69, verified recovery 0.86; median 79 s to diagnose, 394 s to recover. No policy violations, no unnecessary actions.

Finding: the team did not beat one agent here. C1 matched C2 (within the noise of one run on 21 cases) at 60% less cost. C0 is cheapest but took 3 harmful actions: it does no diagnosis.

Source: evaluation report §8.1–8.2 (12 Nov)

RESULTS · E3 AND E4

Memory, budgets and the misleading lesson

E3 · did the playbook help?

42 held-out cases	Memory off	Playbook
Task success	0.57	0.64
Correct action	0.59	0.74
Verified recovery	0.68	0.87
Harmful / unneeded	2 / 1	1 / 0
Cost per case	$0.033	$0.030

Every measure moved the right way, but 3 more successes is within the spread between repeats. No lesson made a case worse.

E4 · tool-call budget

14 cases	20	10	5
Success	0.57	0.71	0.50
Recovery	1.00	1.00	0.73
Cost	$0.033	$0.032	$0.022
Exhausted	1	1	3

10 calls was no worse than 20; 5 cut cost by a third but lowered recovery.

E4 · misleading lesson (D-011)

The old postmortem whose fix would be wrong was **rejected by its preconditions in 2 of 3 runs and used in none**.

0 harmful actions with memory on or off.

But diagnosis took 2.5× longer (186 s vs 75 s) and one run did not recover. 3 runs per arm.

Source: evaluation report §8.3–8.4 (12 Nov)

OPERATING IT

Runs with one command and watches itself

One command

Docker Compose starts the shop, four MCP servers, Prometheus, Grafana and Alertmanager.

Dashboards

Grafana for the shop and the agent: runs, tool calls, cost, blocks, time to recover.

An alert on the agent

CopilotLoopOrBudget fired on a 5-call run and went to the team, not the incident endpoint.

CI eval gate

GitHub Actions runs dev cases and fails below 0.6: the do-nothing agent scored 0.0 and failed.

Tracing

LangSmith, with secrets masked before they leave: 147 and 170 steps on two traced runs.

On-call console

Streamlit: firing alerts, the live plan, the approval screen and the postmortem review.

Source: README; docs/evidence/ (agent-alert, eval-gate, ui)

LIMITS

What these results do not show

One shop, one laptop. Checkout sees 2–4 requests a minute, which slows judging recovery.

Nine fault types. Flag-driven failures the demo ships with, not arbitrary production faults.

An automatic approver. In the experiments it always answers the same way; edit and reject are unit-tested only.

Few repeats. C0, C1 and the budget arms run once on sampled cases; only C2 runs three times, and outputs vary.

Lessons come late. They are shown only after the root cause is chosen: they shape the action, not the diagnosis.

Stand-in retrieval models. The vector index and reranker are lightweight placeholders.

Source: evaluation report §9; docs/playbook

DEMO

Three scenes on the live shop

Scene 1

A payment failure, fixed with approval

The agent finds payment errors in the traces and proposes the flag off; the engineer approves; recovery is verified.

Scene 2

A planted instruction, blocked

A log line saying “restart everything” stays data, and nothing restarts.

Scene 3

An old lesson, rejected

A lesson whose preconditions do not match the live telemetry is rejected (D-011).

BACKUP RECORDING · 12 NOV · ABOUT 20 MIN

github.com/bhavikbpatel/OnCallCopilotG20/releases/tag/demo-2026-11-12

Scene 1 at 0:00 · Scene 2 at 5:15 · Scene 3 at 8:41. Private repo: team and reviewers with access.